with it, for all to see. But the sun does not reach midday at the same moment in Lecce as it does in Venice, or in Florence, or in Turin, because the sun moves from east to west. Midday arrives first in Venice, and significantly later in Turin, and for centuries the clocks in Venice were a good half hour ahead of those in Turin. Every small village had its own peculiar "hour." A train station in Paris kept its own hour, a little behind the rest of the city, as a kind of courtesy toward travelers running late.[39]

In the nineteenth century, the telegraph arrives, trains become commonplace and fast, and the problem arises of properly synchronizing clocks between one city and another. It is awkward to organize train timetables if each station marks time differently. It is in the United States that the first attempt is made to standardize time. Initially, it is proposed to fix a universal hour for the entire world. To call, for instance, "twelve o'clock" the moment at which it is midday *in London*, so that midday would fall at 12 noon in London and around 6 p.m. in New York. The proposal is not well received, because people are attached to local time. In 1883, a compromise is reached with the idea of dividing the world into time zones, thereby standardizing time only within each zone. In this way, the discrepancy between twelve on the clock and local midday is limited to a maximum of about thirty minutes. The proposal is gradually accepted by the rest of the world and clocks begin to be synchronized between different cities.[40]

It can hardly be pure coincidence that, before gaining a university position, the young Einstein worked in the Swiss patent office, dealing specifically with patents relating to the synchronization of clocks at railway stations. It was probably there that it dawned on him: the problem of synchronizing clocks was, ultimately, an insoluble one.

In other words, only a few years passed between the moment at which we agreed to synchronize clocks and the moment at which Einstein realized that it was impossible to do so exactly.

For millennia before clocks, our only regular way of measuring time had been the alternation of day and night. The rhythm of day followed by night also regulates the lives of plants and animals. Diurnal rhythms are ubiquitous in the natural world. They are essential to life, and it seems to me probable that they played a key role in the very origin of life on Earth, since an oscillation is required to set a mechanism in motion. Living organisms are full of clocks of various kinds—molecular, neuronal, chemical, hormonal—each of them more or less in tune with the others.[41] There are chemical mechanisms that keep to a twenty-four-hour rhythm even in the biochemistry of single cells.